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Origami-Folded Sensor Geometries for Cylindrical Tracking Detectors: A Geant4 Comparison

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ABSTRACT: Particle detectors normally use barrel shaped detector. These are usually built using flat silicon tiles, wrapped around a curved structure. However, silicon cannot bend a lot. This forces you to make a choice: either you leave spots with no material (dead zones) or you can use a lot of material and each tile extends a bit over its neighbors (an overlap). Recent designs have mitigated this problem by either tilting the sensor (CMS Collaboration, 2017) itself or by bending the sensor to a small radius and then stitching it (CERN ALICE ITS3). This project considers a different approach, namely the use of different origami structures as detector geometries. This project considers three cylindrically deployable fold families (Cylindrical Miura-ori, Yoshimura, and Kresling) and uses GEANT 4 to answer the following questions:

1. What would be the outcome?
2. Do they have any effect on dead zone or overlap area?
3. What is their performance when compared to a control detector geometry?

Each geometry was modelled parametrically in python and then simulated in GEANT4. The results were then compared to a flat tile barrel of the same radius, which acted as a control. This models Kapton as a continuous structure while silicon is modelled as individual rigid tiles.

Every geometry achieved a higher geometric hit acceptance as compared to the control barrel. However, they also came with an increase in radiation length. Kresling showed the largest acceptance gain (+2.36 percentage points, $p < 10^{-10}$) and had a +5.69% increase in mean X_0. Yoshimura had a smaller acceptance gain (+1.30 percentage points, $p < 10^{-9}$), however, it had the largest increase in radiation length (+8.01% in X_0). Miura-ori had the lowest material penalty, remaining close to the control (+1.88% in X_0). However, it had a modest increase in acceptance (+0.49 percentage points, $p < 10^{-7}$).

This leaves us with a Pareto Frontier rather than a single best geometry. Analyzing the hit distribution reveals the true reason for such a spread: Yoshimura and Kresling concentrate the acceptance gains in localized high hit density regions, regions near overlap seams. This allows them to retain substantial dead-zone fractions (1.86% and 1.45% of (θ, z) bins, respectively) while still increasing hits, all at the cost of increased radiation length. Miura-ori on the other hand has a near uniform hit density throughout and has the lowest dead-zone fraction (0.04%). Multi-parameter sweeps across facet count (N) and fold angle (θ), vertex-position robustness, and multiple-scattering validation against the Highland formula confirm the statistical robustness of these findings.

A parallel study at 50 μm silicon — matching the sensor thickness used by ALICE ITS3 — shows the acceptance advantage of folding is essentially unchanged by sensor thickness, while material traversal drops by a factor of $\sim 4.8\times$ as expected. Compared against ITS3's own material budget at matched physical scope, every geometry studied here, including the flat-tile barrel control, has a higher silicon-only radiation length than ITS3's bent, seamless sensor by roughly 0.01–0.016 percentage points; a naive comparison against ITS3's full engineered per-layer budget (which includes carbon-foam mechanical support this study does not model) would misleadingly suggest otherwise. Folding therefore improves acceptance relative to a flat-tiled barrel at any matched sensor thickness, but does not close the material-budget gap to a continuously bent design.

KEYWORDS: Detector geometry, Particle tracking detectors (Solid-state detectors), Performance of High Energy Physics Detectors, Simulation methods and programs

Contents

1 Introduction

Barrel-shaped particle detectors are normally made using flat silicon tile sensors, constructed around a cylindrical envelope. However, silicon is highly brittle and resists bending due to its mechanical anisotropy. At the same time, the flat tiles force a different choice. Since the flat silicon tiles cannot be wrapped smoothly around a cylindrical envelope, every seam presents a trade off: either we leave a deadzone or we introduce an overlap. Introducing excess material increases Coulomb scattering and degrades vertex resolution.

Detector optimization studies have focused on modifying and optimizing mechanical supports, ultra-light cooling, and high-density interconnects rather than questioning the underlying planar tiling geometry.

Recent detector designs have tried different ways to address this challenge. The ALICE ITS3 upgrade makes use of ultra thin sensors ($\sim 50\ \mu$m), curved to a small angle. These are then stitched together to form the complete barrel. This eliminates the seams inside the sensor layer.

The LHCb Upstream Tracker uses rigid silicon sensors while using a flexible kapton substrate for routing and mechanical packaging.

The CMS Phase-2 Outer Tracker uses a tilted module layout inside the tracker barrel, slanting flat sensor modules along the beam axis to optimize track incidence angles and close acceptance gaps.

While these approaches demonstrate the substantial benefits of non-standard geometries, none explore deployable origami folding to generate the detector surface itself.

Origami Patterns present an unexplored avenue for detector design. These allow a sensor sheet to be tessellated into rigid polygonal facets, interconnected by a kapton substrate shell. This work compares the performance of three origami structures —Miura-ori [1], Yoshimura [2], and Kresling [3]– against a control barrel reference of identical dimensions.

The question is simple: for a fixed cylindrical envelope, does folding reduce dead-zone area, increase radiation-length traversal, or both, relative to tiling? The three fold geometries were generated parametrically in Python and simulated in GEANT4. Each uses the same silicon and Kapton layer stack, so the comparison isolates the effect of geometry rather than material choice. A flat plate was also used to validate the radiation-length scoring.

The radiation-length values reported here cover the modeled sensor and substrate stack only. They are not a complete detector material budget: mechanical supports, cooling, cabling, and readout electronics are not included. The results therefore describe the geometric contribution of folding relative to tiling, rather than the total material of a detector.

2 Methods

2.1 Candidate and Control Geometries

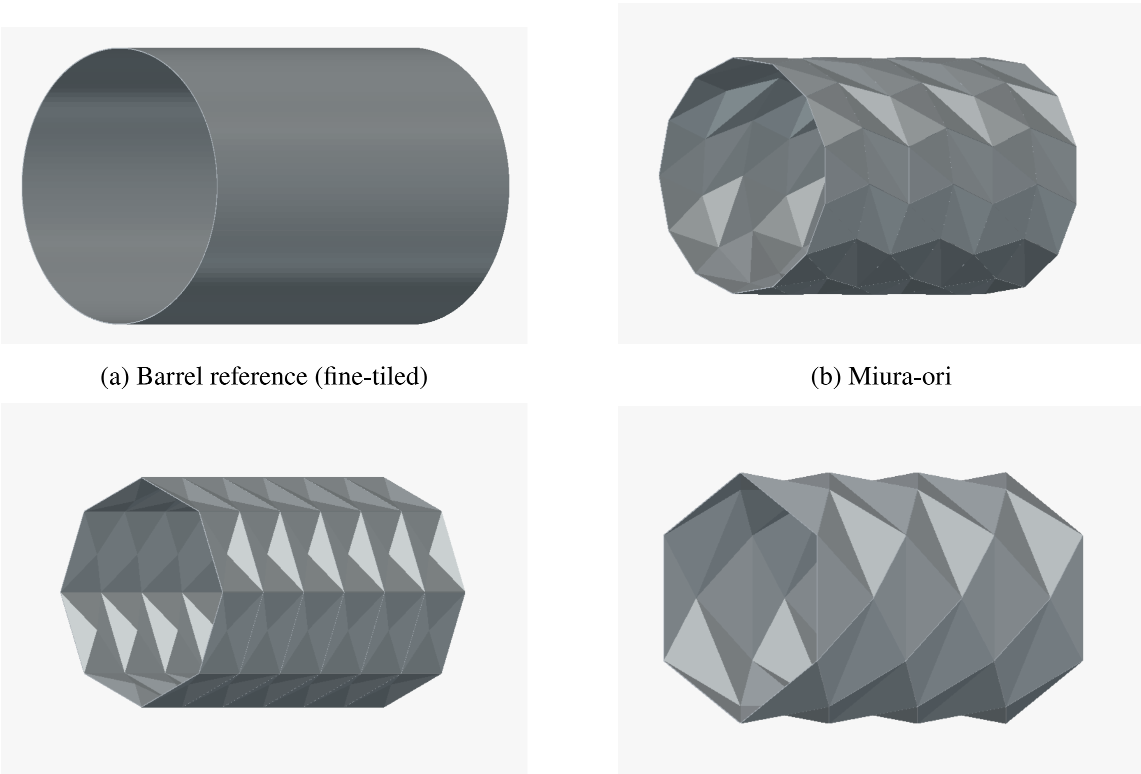
Four cylindrical envelopes sharing an identical outer bounding radius ($R = 40$ mm) and active length ($L = 120$ mm) were constructed and simulated (Figure 1):

- **Barrel reference** — A conventional flat-tile cylindrical barrel serving as the control baseline. Its planar faceted joints establish the baseline dead-zone and overlap performance.

- **Cylindrical Miura-ori** — The classic Miura-ori herringbone parallelogram tessellation [1] mapped onto a cylindrical envelope.

- **Yoshimura** — The diamond-pattern triangular cylindrical tessellation characteristic of thin-shell axial buckling [2].

- **Kresling** — The chiral, triangulated cylinder exhibiting coupled axial compression and rotation [3].



(a) Barrel reference (fine-tiled)

(b) Miura-ori

(c) Yoshimura

(d) Kresling

Figure 1: The four simulated detector geometries in their fully deployed configurations at matched radius (R = 40 mm) and length (L = 120 mm)

Each geometry uses the same two-material layer stack. Silicon is the rigid sensor material (300 μm thickness); however a silicon sensor cannot be bent through a sharp crease without cracking, thus any physically realizable folded sensor sheet requires a flexible backing. Kapton (polyimide, 50 μm thickness) provides this backing, with rigid silicon mounted on the flat facets. This mechanical distinction is part of why these fold patterns are a viable design direction.

Every geometry was built to the same fixed physical envelope and material stack, so comparisons are meaningful. Table 1 summarizes the properties:

Crucially, the silicon and Kapton layers are generated using distinct geometric offset schemes from the underlying polygonal mesh:

1. **Silicon layer (per-facet normal offset):** Each polygonal facet is extruded along its individual face normal and boolean-unioned into a watertight solid. This accurately reflects discrete, rigid sensor tiles whose edges terminate at fold boundaries.

2. **Kapton layer (per-vertex area-weighted normal offset):** The flexible substrate is extruded along shared, vertex-averaged normals, producing a seamless, continuous backing across fold lines without artificial wedge gaps or overlap tearing.

The nominal geometric parameters for each structure are summarized in Table 1. The fine-tiled barrel column count (141 columns) is determined by an inscribed sagitta tolerance of $s \leq 0.01$ mm, ensuring it approximates a smooth cylinder. The fold structures use an axial unit-cell repeat scale of 20 mm.

Table 1: Geometric parameters of the simulated structures at fixed $R = 40$ mm, $L = 120$ mm, 300 μm Si, and 50 μm Kapton.

Geometry	Segmentation / Panel count	Fold parameter
Barrel reference (fine-tiled)	6 rows × 141 cols (846 quads)	—
Barrel reference (coarse-tiled)	12 rows × 24 cols (288 quads)	—
Miura-ori	6 rows × 13 cols	Crease angle $\alpha = 45°$
Yoshimura	6 rings × 8 facets	Fold angle $\theta = 30°$ ($\pi/6$ rad)
Kresling	6 layers × 6 sides	Twist angle $\theta = 30°$/layer

The fine-tiled column count (141) comes from a geometric sagitta tolerance — the maximum allowed deviation of a flat panel from the true 40 mm circle, set to 0.01 mm — chosen so the fine-tiled barrel approximates a smooth cylindrical shell rather than a visibly faceted polygon. We also compare it to a second, coarse-tiled barrel configuration (12 rows × 24 cols, 288 quads, chosen independently of any tolerance criterion) specifically to check how much the barrel's own tile density affects its acceptance and mean path length. That coarse configuration isn't a candidate reference geometry on its own and is just used here.

2.2 Purpose of the Barrel Reference and Baseline Selection

The barrel reference exists to answer a specific question: how much of the dead-zone and radiation-length behavior seen in a fold geometry comes from folding, and how much would be present anyway in any cylindrical detector built at that radius? Without a matched-radius, matched-material control, a fold geometry's dead-zone fraction or radiation-length traversal has no reference point against which it can be judged. A number such as "1.46% of bins are dead zones" only becomes meaningful once it is compared with what a conventional detector at the same radius would show.

The barrel reference was therefore held to the same radius, the same silicon and substrate layer thicknesses, and the same particle beam and event count as the three fold geometries, so that the only geometric variable left is the presence and shape of the fold lines. The barrel is thus the baseline against which the rest of the paper is measured, rather than one option among four being ranked independently.

For the particle simulations, pions with a momentum of 5 GeV/c were chosen. This has three effects:

1. It ensures that the energy loss recorded by the scorer reflects the geometry a track passes through—path length and material—rather than energy-dependent stopping-power effects near the Bragg-like rise at low momentum, which would complicate any comparison between geometries with different effective thicknesses at a given incidence angle.
2. Relativistic effects ensure that pions do not undergo in-flight decay over the required duration.
3. Even considering hadronic interactions with silicon, the simulation explicitly only counts the primary particles towards the acceptance numbers, not secondary particles. This prevents secondary particles from masking geometric gaps in the detector.

2.3 Simulation Pipeline

All geometries are derived from their mathematical formulae and then modeled parametrically in Python. These geometries are then exported to GDML (Geometry Description Markup Language) using explicitly indexed `<tessellated>` solids and parsed directly by Geant4 [4], guaranteeing strict watertightness and preventing navigation track-trapping errors.

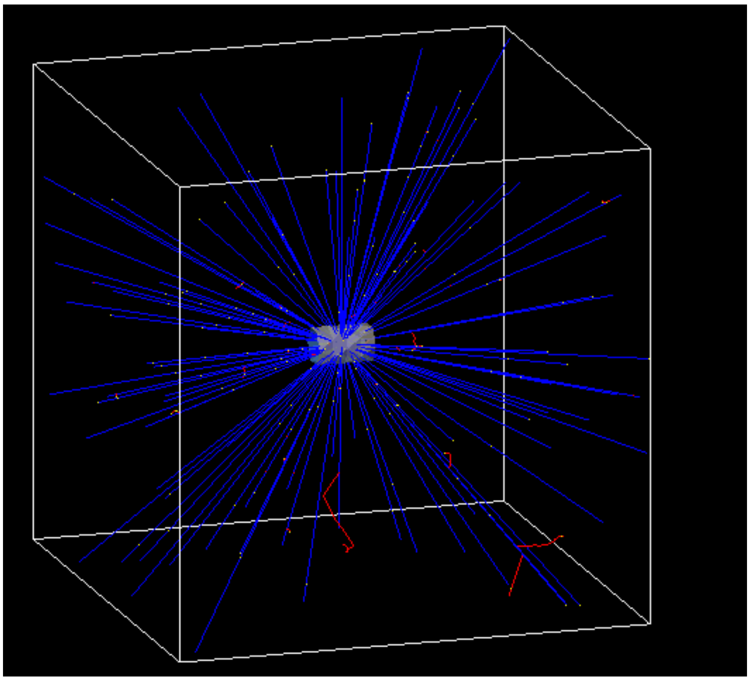


Figure 2: Three-dimensional Geant4 event display showing primary 5 GeV/c π^+ tracks (blue) emitted isotropically from the central interaction vertex through the detector volume, with secondary hadronic tracks indicated in red.

The primary particle generator emits particles isotropically over 4π solid angle. To accurately reflect collider beam-spot distributions, the vertex is sampled from a three-dimensional Gaussian distribution with $\sigma_x = \sigma_y = \sigma_z = 3$ mm centered at the origin (Figure 2). Since isotropic emission over solid angle scales as $\sin\theta$, equatorial polar angle bins ($\theta \approx 90^\circ$) receive greater track density than forward/backward bins; since all geometries share identical source definitions, relative comparisons remain strictly unbiased.

Each run consists of 500,000 primary events per configuration. Step-level sensitive detectors record active volume hits, true traversed 3D path lengths in silicon and Kapton, deposited energy, emission polar angle θ, and local surface incidence angle θ_{local}.

2.4 Evaluation Metrics

Four primary performance metrics are evaluated:

- **Acceptance fraction (A):** The proportion of primary particles producing at least one hit in active silicon ($A = N_{\mathrm{hit}}/N_{\mathrm{total}}$). To ensure clean geometric isolation, all scoring is restricted strictly to primary particle tracks (parentID = 0), filtering out secondary hadronic fragments and delta rays so that both acceptance and multiple-scattering angles reflect the true primary trajectory.

- **Mean and median material traversal in radiation lengths (X/X_0):** Step-level path lengths scaled by material radiation lengths ($X_0(\mathrm{Si}) = 9.366$ cm, $X_0(\mathrm{Kapton}) = 28.575$ cm).

- **Local hit-density enhancement ratio ($R(\theta, z)$):** The per-bin ratio of fold geometry hit density to the barrel reference across a 2D (θ, z) grid.

- **Dead-zone fraction (f_{dead}):** The fraction of (θ, z) phase space bins where the fold hit density falls below the planar baseline ($R(\theta, z) < 1.0$).

2.5 Statistical Treatment

Acceptance fraction is a proportion of a binomial count (hits out of 500,000 trials), so its standard error is computed directly from binomial statistics:

$$SE_A = \sqrt{\frac{A(1-A)}{N}}\,, \tag{2.1}$$

where A is the acceptance fraction and N is the number of primary events. Mean path length in X_0 is likewise reported with a standard error of the mean,

$$SEM = \frac{s}{\sqrt{N_{\mathrm{hit}}}}\,, \tag{2.2}$$

where s is the sample standard deviation of the per-event path length over hit-producing tracks and N_{hit} is the number of hits.

For any pairwise comparison—a fold geometry against the barrel, or against another fold geometry—the two standard errors combine as

$$SE_{\mathrm{diff}} = \sqrt{SE_1^2 + SE_2^2}\,, \tag{2.3}$$

with the corresponding test statistic given by

$$z = \frac{\mathrm{diff}}{SE_{\mathrm{diff}}}\,. \tag{2.4}$$

For acceptance comparisons, diff $= A_1 - A_2$; for path-length comparisons, the corresponding difference in mean path length is used. Whenever two different geometries are compared, a Holm–Bonferroni correction is applied across the relevant family of tests rather than reading each p-value against an uncorrected threshold.

Cohen's h, a standardized effect size for comparing two proportions, is reported alongside z and p wherever a pairwise acceptance comparison is made:

$$h = 2\arcsin\sqrt{A_1} - 2\arcsin\sqrt{A_2}\,. \tag{2.5}$$

A z-score alone conflates effect size with event count. At 500,000 events per geometry, even a small true difference in acceptance fraction can produce a large z-score, whereas h indicates whether that difference is practically meaningful.

This single-run treatment answers one specific question well: could the observed difference be explained by sampling noise within one 500,000-event run? It cannot speak to run-to-run variability from the random seed, or to systematic variation in the simulation pipeline between geometries. To account for this additional variability, each geometry was therefore re-run five times in total, with Welch's two-sample t-test used for pairwise comparisons across the independent runs. Welch's test was chosen because it does not assume that the two samples share a common variance, and the resulting p-values were again Holm–Bonferroni corrected within each test family.

2.6 Radiation-Length and Multiple-Scattering Validation

The Geant4 step-level material accumulator was validated against an analytical planar slab model at normal incidence ($\theta_{\text{local}} = 0°$):

$$(X/X_0)_{\text{Si}} = \frac{0.03\ \text{cm}}{9.366\ \text{cm}} = 3.2031 \times 10^{-3}\,, \tag{2.6}$$

$$(X/X_0)_{\text{Kapton}} = \frac{0.005\ \text{cm}}{28.575\ \text{cm}} = 1.7498 \times 10^{-4}\,, \tag{2.7}$$

$$(X/X_0)_{\text{total}} = 3.2031 \times 10^{-3} + 1.7498 \times 10^{-4} = 3.3781 \times 10^{-3}\,. \tag{2.8}$$

Across five independent validation runs (2,000 events each), the simulated mean path length was $(3.3779 \pm 0.0001) \times 10^{-3}\ X_0$, matching analytical theory to within -0.005% ($z \approx -1.6$, non-significant).

Multiple Coulomb scattering was independently checked against the Highland formula [5, 6]:

$$\theta_0 = \frac{13.6\ \text{MeV}}{\beta c p} z_0 \sqrt{\frac{x}{X_0}} \left[1 + 0.038 \ln\left(\frac{x}{X_0}\right)\right]\,. \tag{2.9}$$

A first attempt, which was just a single application of the formula to the full simulated scattering-angle distribution, gave a puzzling result: $\theta_{0,\text{sim}}/\theta_{0,\text{Highland}} = 0.537 \pm 0.008$ across 20 momentum–geometry configurations ($p \in [0.5, 10.0]$ GeV/c) — a large, consistent shortfall with no physical explanation attached to it.

However, this traced back to the fitting method — i.e., the model chosen — not to failed runs. Since the particles are fired from an isotropic source, around a solid angle of 4π, the resulting scattering angle is not Gaussian. Plotting the probability density of the simulated space-scattering angle for the barrel reference at all five momenta on a log-log scale reveals something interesting: overlaying the corresponding Highland predictions for each momentum shows the two curves moving together at small angle, but after roughly 3–4 times the Highland θ_0, they diverge sharply. The simulated distribution flattens into a power-law tail that extends more than two decades past where the Highland curve has already dropped by several orders of magnitude. A plain unconstrained Gaussian fit to a distribution shaped like this is pulled wide by that tail — exactly the kind of bias a heavy-tailed distribution produces on a naive width estimator, and exactly why the first-pass fit undershot Highland by roughly a factor of two rather than agreeing with it.

Core-plus-tail fit. The scattering-angle distribution was refit as an explicit two-component mixture model, using the 3-D space-scattering angle directly (Section 2.3):

$$p(\theta) = (1 - f) \cdot \mathrm{Rayleigh}(\theta; \sigma) + f \cdot \mathrm{PowerLaw}(\theta; \theta_{\mathrm{min}} = 3\sigma, \alpha)\,, \qquad \theta > 0\,. \tag{2.10}$$

The core is modeled as a Rayleigh distribution rather than a simple Gaussian, which is the correct choice for a space angle: if the two orthogonal plane deflections are each Gaussian with width θ_0 (Highland's own quantity), their magnitude follows a Rayleigh distribution with scale parameter θ_0, with no additional conversion factor required. The tail is modeled in accordance with Rutherford's power law, $dN/d\theta \propto \theta^{-\alpha}$, beyond a tail-onset boundary fixed at three times the fitted core width ($\theta_{\mathrm{min}} = 3\sigma$), tying the tail region to the fit's own core scale rather than to a hand-picked cutoff. All three free parameters (σ, tail fraction f, and tail index α) were fitted simultaneously using unbinned maximum-likelihood estimation against the full per-event scattering-angle sample ($N_{\mathrm{hit}} > 4.1 \times 10^5$ events per configuration). Due to the enormous statistical power of the dataset, formal binned χ^2/dof goodness-of-fit values are sensitive to minute sub-percent level distribution discrepancies (Table 15), while the fitted parameter values ($\sigma/\theta_{0,\mathrm{Highland}} \approx 0.93$, $\alpha \approx 3.0$) demonstrate stable, physically consistent agreement. This is the standard decomposition technique for multiple-scattering distributions in the HEP literature.

Refit this way, $\sigma/\theta_{0,\mathrm{Highland}}$ (the fitted Rayleigh core scale over the Highland prediction) averages 0.930 ± 0.003 across all four geometries and all five momenta (Figure 3, top right; full values in Table 15) — a stable, roughly 7% shortfall, not a factor-of-two one, ranging only from 0.924 to 0.935 across the full 20-point dataset. A weak downward trend with momentum is visible in the data (Pearson $r = -0.42$, $p \approx 0.06$ across the 20 points) but does not reach conventional significance at this sample size, and in any case the entire swept range spans less than a 1.2-percentage-point band (0.924–0.935) — small enough, relative to the ~7% offset itself, that it does not change the interpretation below. Figure 3(b) confirms the underlying shape stability directly for the barrel reference: rescaling both axes by Highland's own θ_0 (angle on the x-axis, density on the y-axis, both in θ_0-natural units) collapses all five momenta onto a single curve, meaning the whole angular distribution — core and tail together — obeys essentially the same shape at every momentum tested. That collapse is itself informative: whatever is producing the residual ~7% gap and the power-law tail is close to a scale-invariant property of the geometry, not a strongly momentum-dependent effect that a small-angle formula would be expected to miss more or less of at different momenta.

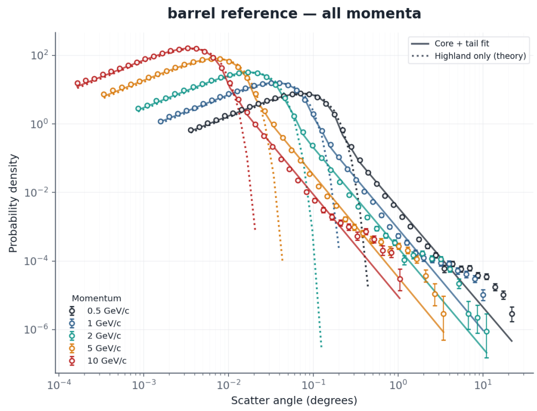
The tail itself matches Rutherford single scattering. The fitted tail power index α averages 2.999 ± 0.046 across all 20 points (Figure 3, bottom right; Table 15) — indistinguishable, to within fit uncertainty, from the value predicted by single Rutherford scattering off individual nuclei. The differential cross section for Rutherford scattering falls as $d\sigma/d\Omega \propto \theta^{-4}$ at small angle; converting from solid angle to a 1-D angular density via the Jacobian $d\Omega = 2\pi \sin\theta\, d\theta \approx 2\pi\theta\, d\theta$ gives $dN/d\theta \propto \theta^{-3}$ — i.e., $\alpha = 3$, matching the fitted tail slope directly, and matching it at every geometry and every momentum tested, not just on average. Despite comprising only 3.3–3.9% of events at any given configuration, this tail dominates the overall variance: Figure 4 shows the tail supplies 86.5% of the total $\sum \theta^2$ on average (range 81.9–89.7%) across the full dataset, for every geometry. A naive RMS-based width estimate is therefore almost entirely a measurement of the rare single-scatter tail, not of the multiple-scattering core Highland's formula describes — the same conclusion reached independently from first principles on the raw per-event data (Appendix A).

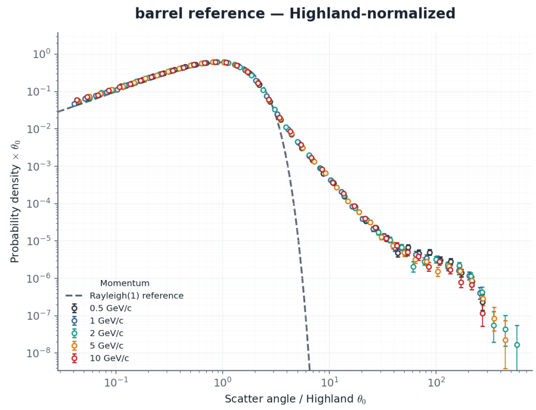
Interpretation. Two effects are cleanly separated by this refit, where the original single-Gaussian comparison conflated them into one confusing number:

1. The Rayleigh **core** — the many-small-deflections regime Highland's formula was derived for — agrees with Highland to within ~7% ($\sigma/\theta_{0,\text{Highland}} = 0.930 \pm 0.003$), consistently across all four geometries and the full momentum range tested. This is the level of agreement expected between a full simulation and Highland's own empirical approximation to it, and is well within Highland's stated accuracy for x/X_0 in the range used throughout this study.

2. A **Rutherford single-scattering tail**, with the theoretically correct power index ($\alpha = 2.999 \pm 0.046$), sits on top of the core, is small in event count (3.3–3.9%) but dominates the raw variance (81.9–89.7% of $\sum \theta^2$), and is a real physical feature of Coulomb scattering that Highland's formula was never designed to capture — Highland is explicitly a small-angle multiple-scattering approximation, not a full-distribution model.

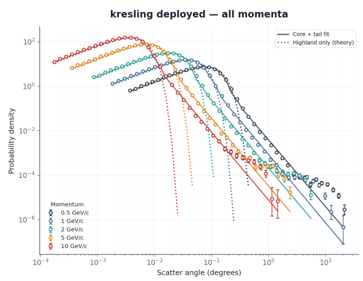
The 0.537 figure from the initial single-Gaussian comparison was therefore a measurement of a fit dominated by tail events it wasn't modeling, not a measurement of core agreement with Highland. Refit with a mixture model that separates the two components, the simulation's core matches Highland's small-angle prediction to within the level of agreement the formula's own derivation would lead one to expect, and the residual structure past the core is physical, not a simulation artifact — its power index is the textbook Rutherford value, its shape is stable across four different geometries, and it collapses onto a single universal curve in Highland-rescaled units across the full momentum range.

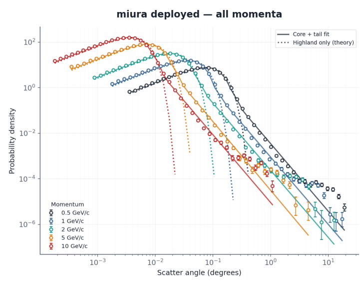


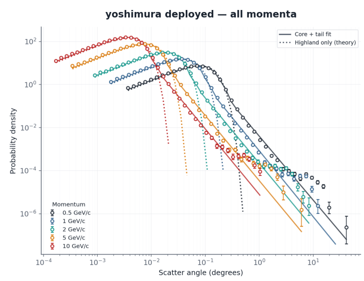


(a) Barrel-reference scattering-angle probability density across all five simulated momenta (0.5–10 GeV/c), mixture fit (solid) vs. Highland-only Rayleigh prediction (dotted)

(b) Barrel-reference data rescaled by each momentum's own Highland θ_0, collapsing all five momenta onto a single curve; dashed line is the Rayleigh(1) reference core shape







(c) Kresling, all momenta

(d) Miura-ori, all momenta

(e) Yoshimura, all momenta

Figure 3: Multiple-scattering validation, Rayleigh-core-plus-Rutherford-tail mixture fit for the barrel reference and candidate fold geometries across all five simulated momenta (0.5–10 GeV/c).

Figure 4: Summary of multiple-scattering mixture fit across all four geometries and all five momenta: fitted core width vs. Highland θ_0 (top left), core-fit/Highland ratio (top right, 0.924–0.935 across the full dataset), tail's share of total $\sum \theta^2$ (bottom left, 82–90% throughout), and fitted tail power index α (bottom right, clustering at the Rutherford-predicted value of 3 across every geometry and momentum).

3 Related Work

There are three main areas of previous work that are related to this project: curved and tilted silicon detectors, origami-based foldable structures, and radiation-length calculations. These areas give the main ideas behind this project, but none of them directly studies the folded detector shapes used here.

3.1 Curved and tilted silicon tracking detectors

The idea of using non-flat detector layers is already being studied in some high-energy physics experiments. One important example is the ALICE ITS3 detector. It uses very thin CMOS silicon sensors that can be bent into a cylindrical shape. The sensors can be bent to radii of 18, 24, and 30 mm, while keeping the material budget at around 0.05% X_0 per layer [7, 8]. Since the sensors are curved, there are no tile joints between separate flat sensors. However, this requires the silicon to be extremely thin and flexible enough to bend without being damaged.

The approach in this project is different. Instead of bending the silicon, the detector is made using flat sections connected by fold lines. This allows the overall detector to have a curved or non-planar shape while keeping the silicon itself flat.

Another useful example is the LHCb Upstream Tracker. It uses normal silicon sensor tiles together with flexible Kapton circuit boards. The Kapton is mainly used for electrical connections, signal routing, and power distribution [9]. The detector stack used in this project is also based on silicon sensors attached to a flexible substrate. However, here the flexible substrate has an additional purpose: it also helps the detector change its overall shape.

The CMS Phase-2 Outer Tracker uses another method. Its detector modules are tilted at different angles along the z-axis. This helps reduce dead areas and improves the detection of tracks, especially for tracks at high pseudorapidity [10]. This is different from the folding method used in this project because the CMS modules are still flat and are simply placed at different angles.

As far as the author knows, these existing detector projects have not directly compared origami-style folded detector geometries with a tiled detector of the same radius using full particle-transport simulation. This is the main area that this project tries to study.

3.2 Origami fold patterns as deployable structures

The fold patterns used in this project are based on existing ideas from origami and deployable structures. These patterns were originally developed for applications such as space structures and thin shells, rather than particle detectors.

Miura-ori is a well-known origami pattern that was developed for folding and deploying large sheets and membranes [1]. One of its main advantages is that the sheet can change from a flat shape to a folded or deployed shape while the individual sections remain mostly rigid. This makes it useful when a large change in shape is needed without bending the whole material.

The Yoshimura pattern comes from the study of thin cylindrical shells under compression. It has a diamond-like folding pattern and was later used as a deliberate folding structure [2]. Kresling is another fold pattern that produces both rotation and shortening when it is compressed. It can also have different stable folded and deployed states [3].

These origami patterns are therefore not new. The new part of this project is using them for a detector and checking whether they can provide good hit acceptance while keeping the amount of material crossed by particles reasonably low.

3.3 Radiation length and material-budget simulation

Radiation length is important when studying the amount of material in a particle detector. For a material with thickness t, the material traversed at normal incidence can be written as

$$\frac{X}{X_0} = \frac{t}{X_0}. \tag{3.1}$$

If a particle crosses the material at an angle θ_{local}, it travels a longer distance through the material. The path length can be approximately written as

$$t_{\text{path}} = \frac{t}{\cos\left(\theta_{\text{local}}\right)}. \tag{3.2}$$

These equations are used to check whether the material-budget calculations in the simulation are working correctly.

GEANT4 [4] is commonly used in particle-physics simulations. It can simulate particle transport, energy loss, scattering, and interactions with detector materials. In this project, GEANT4 is used to compare the different folded detector geometries.

The sensor and substrate materials are kept the same for all the detector designs. Only the geometry and fold pattern are changed. This makes it possible to see whether differences in hit acceptance and material traversal are mainly caused by the detector shape.

4 Results

4.1 Barrel Tile-Count Sensitivity and Headline Performance

Before any results can be read, it is important to understand the sensitivity of barrel baseline based on number of tiles used to form it (N_{cols}). An inscribed n-sided polygon of radius R exhibits a geometric sagitta $s = R(1 - \cos(\pi/n))$. Coarse faceting ($N_{\text{cols}} = 24$, $s = 0.342$ mm) increases the mean local incidence angle by 1.70° relative to fine faceting ($N_{\text{cols}} = 141$, $s = 0.0099$ mm), shifting acceptance by +0.512 pp and mean path length by +1.54% ($z \approx 6.9$, Table 2). This is why the high-density fine-tiled barrel ($N_{\text{cols}} = 141$) is employed as the strict reference baseline throughout the subsequent differentiated-material evaluations.

Table 2: Barrel baseline sensitivity to tile faceting (500,000 primary pions, $R = 40$ mm).

Configuration	Hits / Total	Acceptance	Mean Path (X_0)	$\langle\theta_{\text{local}}\rangle$
Barrel (coarse, 24 cols)	418,388 / 500,000	0.836776	$0.0037944 \pm 1.17 \times 10^{-6}$	27.97°
Barrel (fine, 141 cols)	415,828 / 500,000	0.831656	$0.0037367 \pm 1.14 \times 10^{-6}$	26.27°

Table 3 reports the single-run headline results under the differentiated Kapton mesh and 3 mm vertex smear. All three origami geometries demonstrate statistically significant acceptance gains over the fine-tiled barrel, accompanied by increases in radiation-length traversal.

Table 3: Single-run exploratory performance metrics for candidate geometries (500,000 primaries each). Multi-run replication ground-truth means and uncertainties are reported in Table 5.

Geometry	Hits / Total	Acceptance	Mean X_0	95% CI (X_0)	Median X_0	IQR (X_0)
Barrel (fine)	415,709 / 500,000	0.831418	0.003977	[0.003975, 0.003979]	0.003713	0.000860
Miura-ori	418,159 / 500,000	0.836318	0.004053	[0.004050, 0.004055]	0.003766	0.000870
Kresling	427,662 / 500,000	0.855324	0.004204	[0.004201, 0.004208]	0.003788	0.000827
Yoshimura	422,427 / 500,000	0.844854	0.004295	[0.004291, 0.004299]	0.003950	0.000932

Pairwise hypothesis tests across all six geometric combinations (Table 4) confirm that Kresling achieves the highest overall acceptance (+2.362 pp over barrel, $t = -128.89$, $d = -81.52$), followed by Yoshimura (+1.304 pp, $t = -47.22$, $d = -29.86$) and Miura-ori (+0.493 pp, $t = -23.44$, $d = -14.82$). In material cost, Yoshimura incurs the largest penalty (+8.01% over barrel), exceeding Kresling (+5.69%) and Miura-ori (+1.88%).

Table 4: Pairwise acceptance comparisons across 5-run replications (Welch's two-sample t-test, Holm–Bonferroni corrected).

Pairwise comparison	ΔA (pp)	t-statistic	p-value	Holm thresh.	Sig.?	Cohen's d
Barrel vs. Kresling	−2.362	−128.89	$\sim 10^{-11}$	0.0083	Yes	−81.52
Miura-ori vs. Kresling	−1.868	−119.98	$\sim 10^{-12}$	0.0100	Yes	−75.88
Barrel vs. Yoshimura	−1.304	−47.22	$\sim 10^{-10}$	0.0125	Yes	−29.86
Kresling vs. Yoshimura	+1.058	+44.66	$\sim 10^{-8}$	0.0167	Yes	+28.24
Miura-ori vs. Yoshimura	−0.810	−31.32	$\sim 10^{-8}$	0.0250	Yes	−19.81
Barrel vs. Miura-ori	−0.493	−23.44	$\sim 10^{-8}$	0.0500	Yes	−14.82

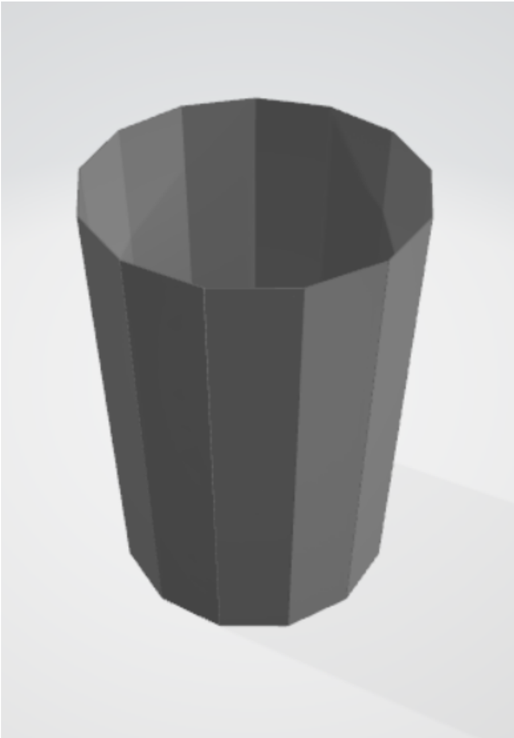
The underlying run-to-run standard deviations ($SD_A \leq 0.05$ pp, Table 5) across five independent 500,000-event runs per geometry are an order of magnitude smaller than the inter-geometry acceptance margins, which is what makes the Welch's t-test results above (Table 4) decisive rather than borderline.

Table 5: Five-run replication summary (Mean ± SD and SEM, $5 \times 500{,}000$ primaries).

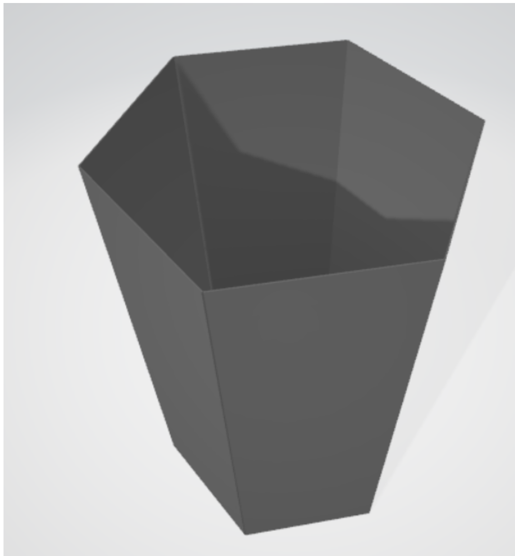
Geometry	Mean A	SD_A	SEM_A	Mean X_0	SD_{X0}	SEM_{X0}
Barrel (fine)	0.831784	0.000366	0.000164	0.0039765	5.36×10^{-7}	2.40×10^{-7}
Miura-ori	0.836718	0.000296	0.000132	0.0040514	1.13×10^{-6}	5.07×10^{-7}
Kresling	0.855400	0.000184	0.000082	0.0042029	1.66×10^{-6}	7.40×10^{-7}
Yoshimura	0.844820	0.000497	0.000222	0.0042950	4.77×10^{-7}	2.13×10^{-7}

4.2 Circumferential Facet-Count (N) Sweep and Geometric Singularities

Sweeping facet count $N \in \{6, 8, 12, 16, 24, 32\}$ reveals fundamental convergence trends (Table 6 and Figure 6). For all three families, acceptance gains asymptotically converge toward the smooth barrel limit as N increases (Kresling drops from +2.31 pp at $N = 6$ to −0.00 pp at $N = 32$).



(a) Degenerate Kresling (N = 12, θ_{twist} = 30°)



(b) Degenerate Yoshimura (N = 6, θ = 30°)

Figure 5: Direct 3D renders of the excluded degenerate configurations resulting from closed-form rotational periodicity cancellation: (a) Kresling N = 12 collapsing to an untwisted faceted cylinder, and (b) Yoshimura N = 6 collapsing to a plain tapered hexagonal frustum.

Table 6: Facet-count (N) sweep differences relative to the fine barrel reference.

N	Kresling ΔA	Kresling ΔX_0	Miura ΔA	Miura ΔX_0	Yoshi ΔA	Yoshi ΔX_0
6	+2.31 pp	+5.72%	+2.34 pp	+14.91%	— (excl.)	— (excl.)
8	+1.26 pp	+2.49%	+1.39 pp	+6.41%	+1.34 pp	+8.03%
12	— (excl.)	— (excl.)	+0.56 pp	+2.30%	+0.60 pp	+13.24%
16	+0.31 pp	+1.70%	+0.41 pp	+1.23%	+0.38 pp	+18.39%
24	+0.09 pp	+2.26%	+0.17 pp	+0.56%	+0.17 pp	+29.53%
32	−0.00 pp	+2.57%	+0.13 pp	+0.31%	+0.14 pp	+41.78%

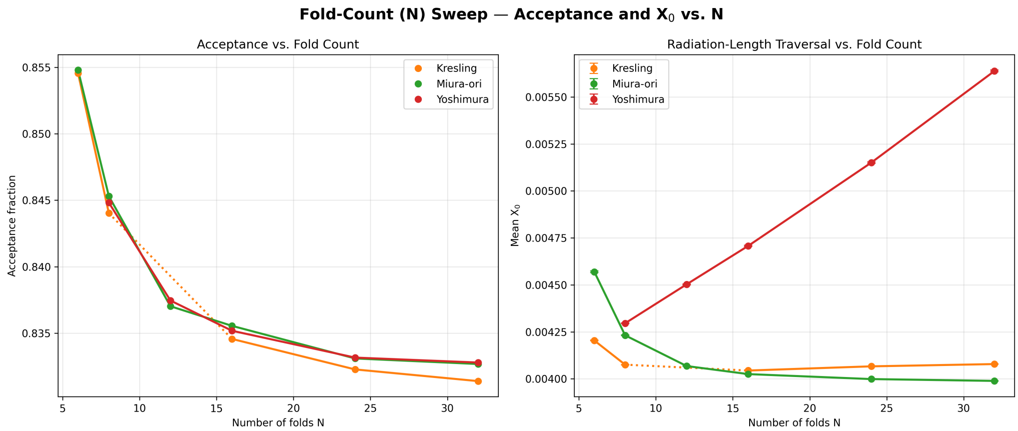


Figure 6: Acceptance fraction gain ΔA (left) and mean radiation-length traversal excess ΔX_0 (right) as a function of circumferential facet count N across candidate fold families relative to the fine-tiled barrel baseline.

In contrast, radiation length diverges between families: Miura-ori exhibits a monotonically decreasing material excess with larger N (falling to +0.31% at $N = 32$), whereas Kresling plateaus into a narrow band between +1.70% and +2.57% for $N \geq 8$, after an initial +5.72% penalty at the smallest facet count ($N = 6$), and Yoshimura exhibits an escalating material penalty, surging from +8.03% at $N = 8$ to +41.78% at $N = 32$. The nominal Miura-ori configuration ($N = 13$) utilizes an odd facet count to satisfy circumferential anti-symmetric herringbone seam closure, interpolating smoothly between the $N = 12$ and $N = 16$ swept points. Two singular configurations—Yoshimura $N = 6$ ($\theta = \pi/N$) and Kresling $N = 12$ ($\theta_{\text{twist}} = 360°/N$)—collapse into untwisted prismatic polygons (Figure 5) and are formally excluded.

4.3 Fold-Angle (θ) Sensitivity

Varying the fold angle $\theta \in \{15°, 30°, 45°, 60°, 75°\}$ illustrates contrasting geometric stability (Table 7 and Figure 7). Miura-ori and Kresling remain virtually flat across the entire angular domain (acceptance varying by < 0.15 pp). Conversely, Yoshimura exhibits extreme non-linear growth: acceptance increases from 0.8448 to 0.9415 while mean X_0 triples from 0.004060 to 0.013303 X_0, driven by steep facet inclination angles. Kresling $\theta = 60°$ represents an exact rotational periodicity degeneracy and is excluded.

Table 7: Fold-angle (θ) sweep performance across fold families (500,000 primaries each).

θ	Kresling A	Kresling X_0	Miura A	Miura X_0	Yoshi A	Yoshi X_0
15°	0.8546	0.004184	0.8367	0.004076	0.8448	0.004060
30°	0.8554	0.004201	0.8369	0.004065	0.8450	0.004295
45°	0.8547	0.004181	0.8373	0.004053	0.8552	0.005522
60°	— (excl.)	— (excl.)	0.8371	0.004039	0.8918	0.008369
75°	0.8554	0.004575	0.8362	0.004031	0.9415	0.013303

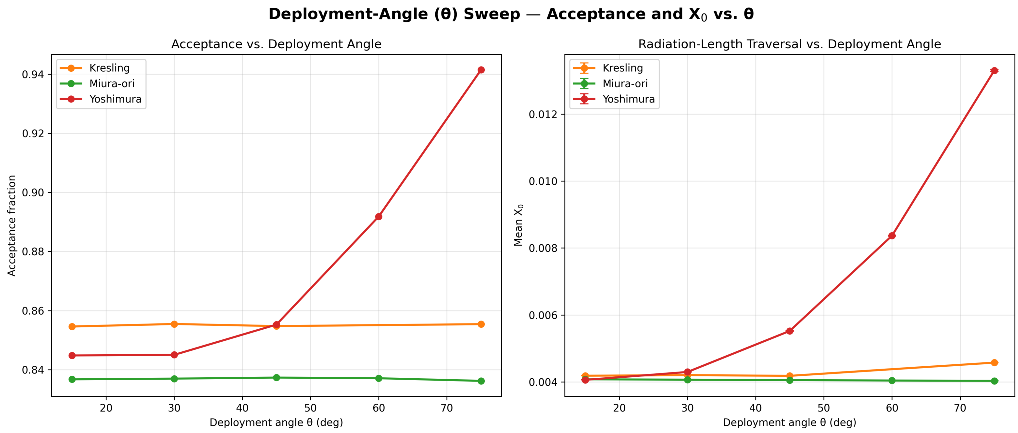


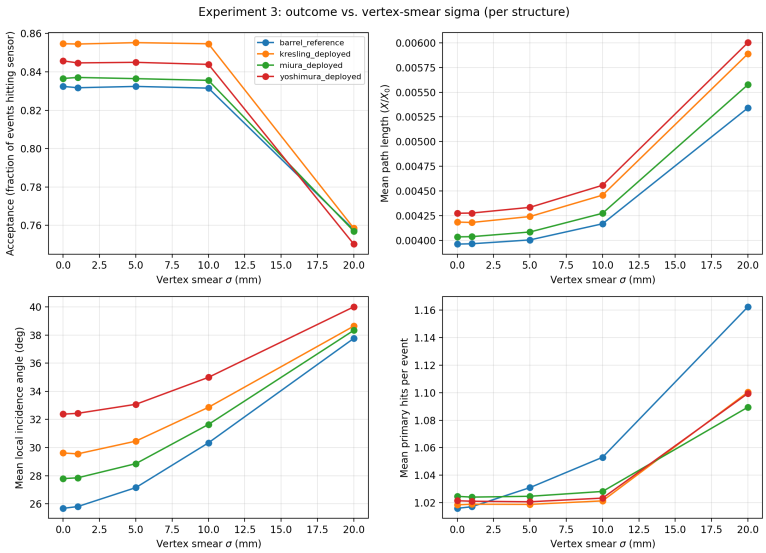
Figure 7: Acceptance fraction and mean radiation-length traversal X_0 versus fold angle θ across candidate fold families.

4.4 Vertex-Position Robustness

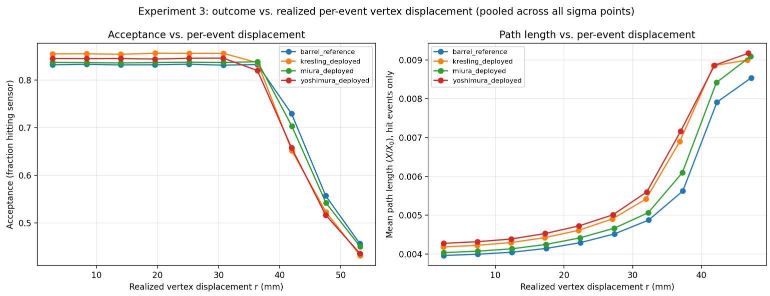
To confirm that the 3 mm vertex smear does not distort results, the Gaussian smear parameter was swept over $\sigma \in [0, 20]$ mm alongside radial vertex displacement tests (Table 8 and Figure 8). Performance is entirely stable up to $\sigma = 10$ mm ($|\Delta A| < 0.2$ pp). At $\sigma = 20$ mm, the beam spot becomes comparable to the detector radius (40 mm), causing substantial off-center track leakage.

Table 8: Acceptance and mean X_0 as a function of 3D Gaussian vertex smear σ.

σ (mm)	Barrel A	Barrel X_0	Kresling A	Kresling X_0	Miura A	Miura X_0	Yoshi A	Yoshi X_0
0	0.8324	0.003962	0.8546	0.004185	0.8365	0.004035	0.8457	0.004274
1	0.8317	0.003965	0.8545	0.004181	0.8371	0.004037	0.8447	0.004276
5	0.8324	0.004004	0.8552	0.004241	0.8364	0.004084	0.8449	0.004334
10	0.8315	0.004168	0.8546	0.004458	0.8355	0.004274	0.8439	0.004558
20	0.7577	0.005341	0.7585	0.005888	0.7570	0.005576	0.7503	0.006002



(a) Acceptance and X_0 vs. 3D Gaussian vertex smear width σ



(b) Acceptance and X_0 vs. systematic radial vertex displacement

Figure 8: Vertex-position robustness evaluations: (a) Gaussian smear width $\sigma \in [0, 20]$ mm, and (b) systematic static radial displacement across geometries.

4.5 Dead-Zone Fractions and Multi-Objective Pareto Frontier

Table 9 reports the dead-zone fraction (f_{dead}) across five independent runs. The dead-zone ranking is inversely correlated with acceptance: Miura-ori exhibits negligible dead zones (0.042%), whereas Kresling (1.451%) and Yoshimura (1.861%) retain significant under-dense areas. Figure 9 illustrates the dead-zone fraction across detection threshold sweeps.

Table 9: Dead-zone fraction (f_{dead}) across five replication runs per fold geometry.

Structure	Run A	Run B	Run C	Run D	Run E	Mean	SD	SEM
Miura-ori	0.069%	0.069%	0.000%	0.069%	0.000%	0.041%	0.038%	0.017%
Kresling	1.319%	1.597%	1.389%	1.667%	1.285%	1.451%	0.171%	0.076%
Yoshimura	1.771%	1.840%	1.632%	1.944%	2.118%	1.861%	0.183%	0.082%

While the table shows the deadzone data for a specified threshold, the figure below shows the different values for deadzone ratio across a sweep of different threshold values, ranging from 0 to 1.

And, in this case, we can see that the order remains the same, until the very end wherein Kresling takes a small advantage. However, at that point, the advantage is negligible and irrelevant.

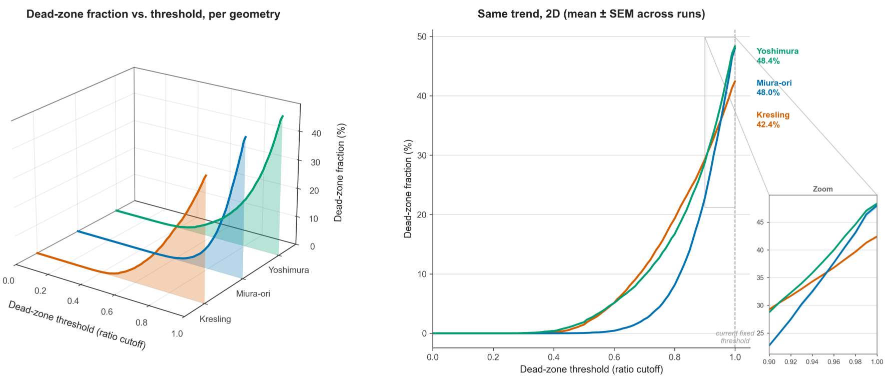
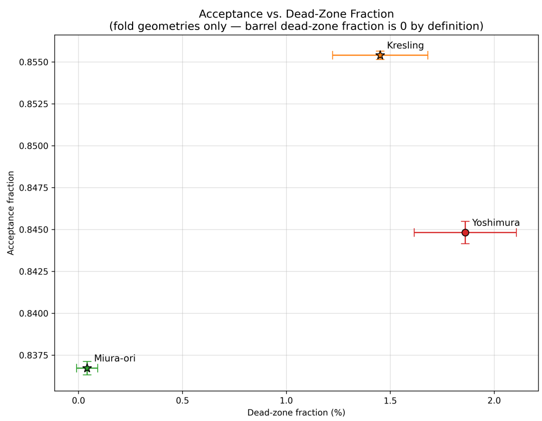
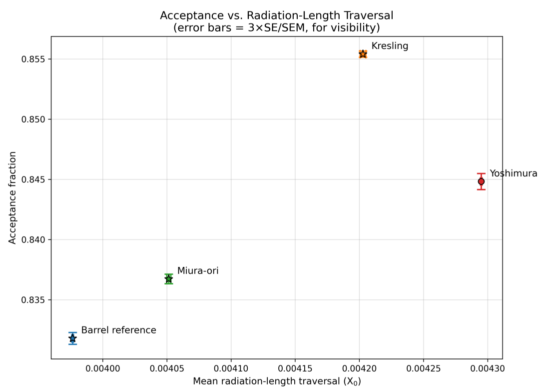


Figure 9: Dead-zone fraction sensitivity across detection threshold parameters for candidate fold geometries, showing the 3D response surface (left) and 2D mean ± SEM comparison (right).



(a) Acceptance fraction vs. dead-zone fraction



(b) Acceptance fraction vs. mean X_0

Figure 10: Pairwise 2D objective trade-off projections for acceptance: (a) acceptance fraction vs. dead-zone fraction, and (b) acceptance fraction vs. mean X_0.

Figures 10–13 demonstrate that the four geometries constitute a classic Pareto set (quoted performance figures correspond to the 5-run replication means in Table 5):

- **Kresling** maximizes acceptance ($A = 85.54\%$) at intermediate material cost ($X/X_0 = 4.20 \times 10^{-3}$).
- **Miura-ori** minimizes material penalty ($X/X_0 = 4.05 \times 10^{-3}$) and nearly eliminates dead zones ($f_{\text{dead}} = 0.04\%$) with modest acceptance gain ($A = 83.67\%$).
- **Yoshimura** is Pareto-dominated, incurring both the highest material cost ($X/X_0 = 4.30 \times 10^{-3}$) and highest dead-zone fraction (1.86%) for inferior acceptance (84.48%).

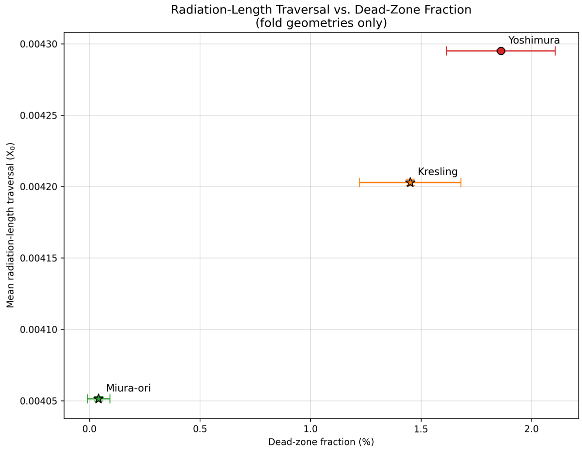


Figure 11: Mean radiation-length traversal X_0 versus dead-zone fraction across candidate fold geometries.

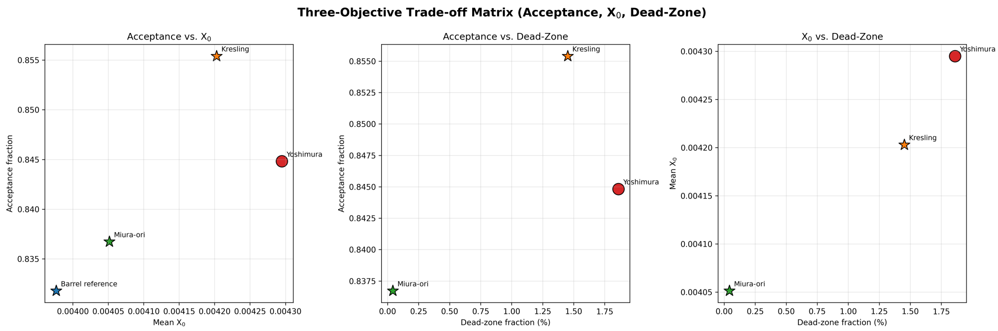


Figure 12: Comprehensive three-panel trade-off matrix pairing acceptance fraction, mean X_0, and dead-zone fraction for all geometries.

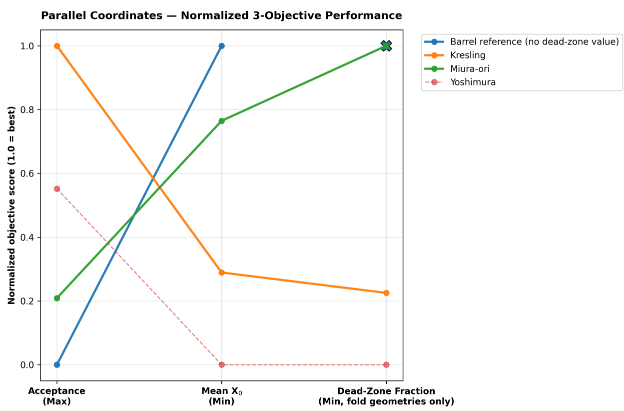


Figure 13: Parallel coordinate representation across normalized performance objectives (where upward orientation denotes superior performance).

4.6 Spatial Structure of Dead Zones and Overlaps

Two-dimensional (θ, z) hit-density and material maps (Figures 14–16) elucidate the spatial mechanism:

- **Miura-ori (Figure 14):** Displays a smooth, quasi-uniform hit-density profile ($R \approx 1$) and continuous material gradient, matching its low dead-zone fraction.
- **Yoshimura (Figure 15):** Shows sharp periodic diamond bands where overlap seams ($R > 1.5$) alternate with localized dead-zone valleys ($R < 1$).
- **Kresling (Figure 16):** Exhibits helical band structures with localized material peaks at fold vertices.

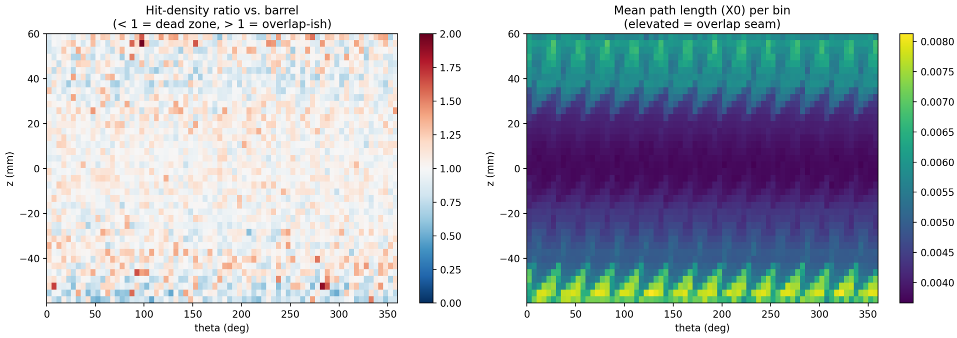


Figure 14: Spatial hit-density enhancement ratio $R(\theta, z)$ (left) and mean X_0 (right) for Cylindrical Miura-ori relative to the barrel control.

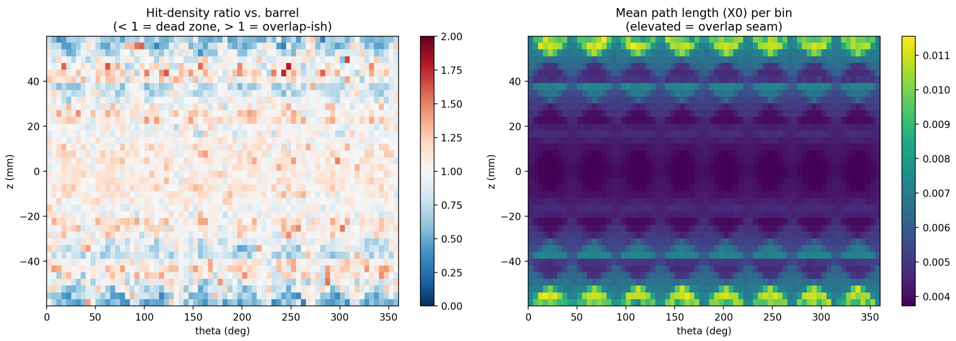


Figure 15: Spatial hit-density enhancement ratio $R(\theta, z)$ (left) and mean X_0 (right) for Yoshimura relative to the barrel control.

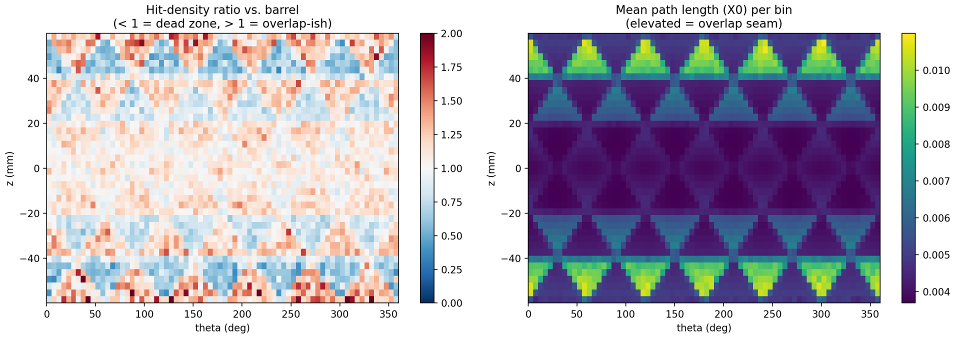


Figure 16: Spatial hit-density enhancement ratio $R(\theta, z)$ (left) and mean X_0 (right) for Kresling relative to the barrel control.

4.7 Path Length vs. Local Incidence Angle

Binned path lengths as a function of local incidence angle θ_{local} (Table 10) confirm textbook $1/\cos(\theta_{\text{local}})$ scaling near normal incidence (5°–25°), where all geometries match within 0.3%. At grazing angles ($\geq 55°$), fold-line seam traversals induce severe non-linear path length growth, with Kresling exceeding the planar barrel baseline by +27.1% at 65° (0.00796 X_0 vs. 0.00627 X_0), while origami folds maintain active hit coverage out to 75°–85° where the flat barrel ceases to capture tracks.

Table 10: Mean path length in X_0 across local incidence angle bins θ_{local}.

θ_{local}	Barrel X_0	Miura-ori X_0	Kresling X_0	Yoshimura X_0
5°	0.003384	0.003392	0.003382	0.003390
15°	0.003480	0.003481	0.003478	0.003487
25°	0.003710	0.003704	0.003708	0.003708
35°	0.004105	0.004095	0.004061	0.004089
45°	0.004750	0.004747	0.004692	0.004642
55°	0.005578	0.005639	0.005599	0.005898
65°	0.006268	0.006973	0.007964	0.007408
75°	—	0.007162	0.010032	0.009871
85°	—	—	—	0.009458

4.8 Thin-Silicon Variant (50 μm) and Comparison to ITS3

The results in Sections 4.1–4.7 use the nominal 300 μm silicon thickness given in Table 1. To check how the results change for a thinner sensor, all four geometries were also simulated with the silicon thickness reduced to 50 μm. This thickness is comparable to the 40–50 μm silicon used for the ALICE ITS3 sensors [7, 8].

For this comparison, the radius, active length, Kapton thickness, fold parameters, particle beam, event count, and vertex smear were kept the same as the nominal configuration. This means that the main change between the two cases is the silicon thickness.

Each geometry was simulated using a single 500,000-event run. This follows the single-run treatment used for the headline results in Table 3 rather than the five-run replication used in Tables 5 and 9. The acceptance comparisons therefore use the binomial standard errors and z-tests described in Section 2.5, rather than the Welch's t-test used for the five-run results. Since only one run was performed at 50 μm, run-to-run variation at this thickness has not been measured.

4.8.1 Headline performance at 50 μm

Table 11 shows the acceptance and mean radiation-length traversal for the four geometries with 50 μm silicon.

Table 11: Single-run headline performance at 50 μm silicon (500,000 primaries each).

Geometry	Hits / Total	Acceptance	Mean X/X_0	95% CI	Median X/X_0	IQR
Barrel (fine)	415,826 / 500,000	0.831652	0.000840	[0.000839, 0.000840]	0.000782	0.000182
Miura-ori	418,769 / 500,000	0.837538	0.000850	[0.000849, 0.000850]	0.000786	0.000183
Kresling	427,686 / 500,000	0.855372	0.000868	[0.000867, 0.000869]	0.000777	0.000175
Yoshimura	422,135 / 500,000	0.844270	0.000894	[0.000893, 0.000895]	0.000817	0.000193

All three origami geometries again give higher acceptance than the fine-tiled barrel. Kresling gives the largest increase, followed by Yoshimura and Miura-ori.

The pairwise comparisons in Table 12 show that all three acceptance gains remain statistically significant.

Table 12: Pairwise acceptance and material-cost comparisons vs. barrel, 50 μm silicon (single run, z-test per Eqs. 2.1–2.4). Following Section 2.5, Holm–Bonferroni correction is applied across the three comparisons.

Comparison	ΔA (pp)	z	p	Holm thresh.	Sig.?	Cohen's h	ΔX_0 (%)
Barrel vs. Kresling	+2.372	32.66	$< 10^{-16}$	0.0167	Yes	+0.065	+3.38%
Barrel vs. Yoshimura	+1.262	17.12	$< 10^{-16}$	0.025	Yes	+0.034	+6.49%
Barrel vs. Miura-ori	+0.589	7.92	2.4×10^{-15}	0.05	Yes	+0.016	+1.21%

The ordering is therefore the same as at 300 μm. Kresling provides the largest acceptance gain, Yoshimura provides the second largest, and Miura-ori provides the smallest gain. The material penalty is also still smallest for Miura-ori.

4.8.2 Effect of silicon thickness on acceptance

Table 13 compares the acceptance gain over the barrel at 300 μm and 50 μm silicon.

The acceptance gains are very similar at the two silicon thicknesses. Kresling changes by only 0.010 percentage points, Yoshimura by −0.042 percentage points, and Miura-ori by +0.096 percentage points.

These differences are small compared with the run-to-run spread measured for the 300 μm simulations in Table 5, where $\mathrm{SD}_A \leq 0.05$ pp. This suggests that the acceptance difference between the geometries is mainly controlled by the geometry of the facets, seams, and dead zones rather than by the silicon thickness.

Table 13: Acceptance gain over barrel, 300 μm vs. 50 μm silicon.

Geometry	ΔA at 300 μm (pp)	ΔA at 50 μm (pp)
Miura-ori	+0.493	+0.589
Kresling	+2.362	+2.372
Yoshimura	+1.304	+1.262

However, this should not be treated as a full statistical demonstration of thickness independence. Only one run was performed at 50 μm. A five-run replication at 50 μm would be needed to measure the run-to-run variation directly.

The material traversal changes much more strongly when the silicon thickness is reduced. The mean X/X_0 decreases by approximately 4.74–4.84× across the four geometries. The reduction is smaller than the factor of six expected from the silicon thickness alone. This is because the Kapton layer remains at 50 μm in both simulations. At the lower silicon thickness, the Kapton therefore makes up a larger fraction of the total material traversal.

4.8.3 Comparison to ALICE ITS3

The ALICE ITS3 detector provides a useful reference for the 50 μm case. However, two different material-budget numbers are used for ITS3 and they describe different parts of the detector.

The silicon sensor itself contributes approximately 0.053% X_0. The full engineered layer, including the silicon together with carbon-foam stiffeners, structural glue, and air cooling, is approximately 0.09% X_0 [7, 8].

These two values should therefore not be treated as two estimates of the same quantity. The 0.053% X_0 value describes the silicon only, while the 0.09% X_0 value describes a larger part of the detector structure.

The simulation in this project contains silicon and Kapton only. Table 14 therefore compares the silicon part of the simulation with the ITS3 silicon-only value, and the complete simulated silicon + Kapton stack with the full ITS3 layer budget.

Table 14: Mean X/X_0 at 50 μm silicon compared with the two ITS3 reference values.

Geometry	Si-only X/X_0	Difference from ITS3 Si-only (0.053%)	Si+Kapton X/X_0	Difference from ITS3 full budget (0.09%)
Barrel (fine)	0.0633%	+0.0103 pp	0.0840%	−0.0060 pp
Miura-ori	0.0646%	+0.0116 pp	0.0850%	−0.0050 pp
Kresling	0.0674%	+0.0144 pp	0.0868%	−0.0032 pp
Yoshimura	0.0688%	+0.0158 pp	0.0894%	−0.0006 pp

The silicon-only comparison is the more direct comparison because both values describe the same physical component. All four geometries have a higher silicon-only material traversal than the ITS3 value. The difference ranges from approximately 0.010 to 0.016 percentage points.

This difference is mainly related to the way the sensor surface is constructed. ITS3 uses very thin silicon that can be bent into a cylindrical shape. A track crossing the sensor therefore sees a surface that follows the cylindrical envelope more closely. The geometries in this project instead use

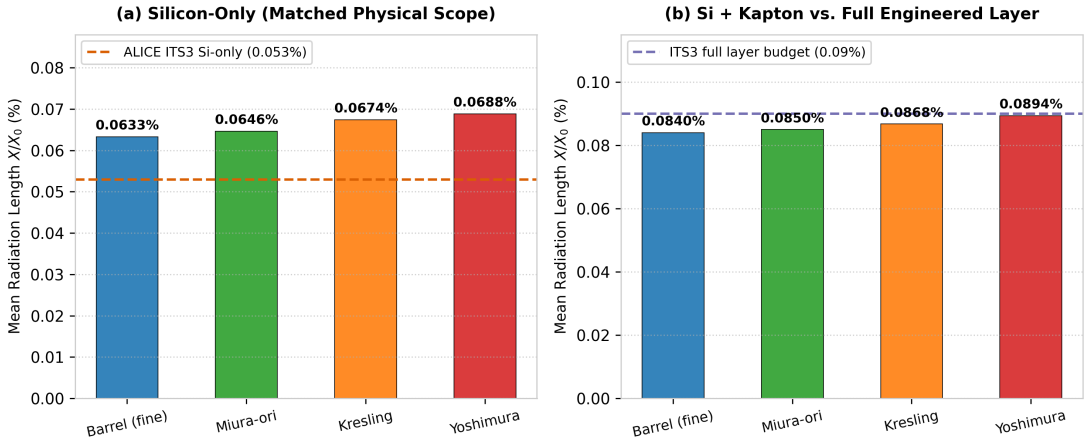


Figure 17: Comparison of mean radiation length X/X_0 at 50 μm silicon against ALICE ITS3 benchmark references [7, 8]: (a) silicon-only traversal evaluated at matched physical scope against the ITS3 bent sensor baseline (0.053% X_0), and (b) simulated silicon + Kapton stack compared with the full engineered ITS3 layer budget (0.09% X_0, which includes carbon-foam supports, glue, and cooling not modeled here).

flat silicon facets at a fixed radius. Tracks that do not cross these facets close to normal incidence therefore travel through a longer silicon path.

This effect is also seen in the 300 μm results in Section 4.7, where path length increases with local incidence angle. The fold geometries do not remove this effect. Yoshimura shows the largest silicon-only material traversal of the four geometries, consistent with its larger facet inclination angles.

The second comparison, using silicon plus Kapton, gives a different result. All four simulated geometries are below the 0.09% X_0 ITS3 full-layer value. However, this is not a direct material-budget comparison.

The ITS3 full-layer value includes structural components such as carbon-foam stiffeners and structural glue, while the present simulation includes only a 50 μm Kapton backing. Mechanical supports, glue, cooling, cabling, and readout electronics are not included in this model, as described in Sections 1 and 6. The Kapton layer is therefore a much simpler and lighter representation of the detector structure.

The fact that the simulated silicon + Kapton values are below 0.09% X_0 should therefore not be interpreted as showing that the folded geometries have a lower material budget than ITS3. The two simulations do not contain the same structural components.

Overall, the 50 μm comparison shows that reducing the silicon thickness substantially reduces the material traversal while leaving the acceptance advantage of the folded geometries almost unchanged. The folded geometries do not, however, achieve a lower silicon-only material traversal than the ITS3 sensor.

The main result of the folding approach therefore remains the same as in the 300 μm study. Folding can increase geometric acceptance relative to a flat-tiled barrel, but this comes with an increase in material traversal. At 50 μm, Kresling again gives the largest acceptance gain, while Miura-ori remains the closest to the barrel in material cost. The comparison with ITS3 also shows the limitation of this approach: a bent, seamless sensor can reduce the path-length penalty that remains when a cylindrical surface is approximated using flat silicon facets.

5 Discussion

Based on the results above, it is clear that there is no single geometry that beats all others, rather, the result is in the form of a pareto frontier. The core finding of this paper is that folding increases the hit acceptance in detector geometries but it also leads to an increase in the radiation length when compared to a flat tile barrel of matching radius. However, the differences are spread out amongst the three families. It also shows that acceptance gain in Yoshimura and Kresling is concentrated in overlap regions, similarly, these geometries are also the ones which have the highest dead zone fractions.

Miura-ori is closest to the barrel's radiation-length (+1.88%) but it also shows the smallest acceptance gain (+0.49 percentage points). However, the gain is still real, as shown by the multi-run replication statistics ($t = -23.44$, $p < 10^{-7}$). It is also the geometry with the smallest dead zone fractions, matching the barrel's dead-zone pattern closely, rather than improving on it.

Kresling geometry has the largest acceptance gain over the normal barrel (+2.36 percentage points, $t = -128.89$), even when compared to other geometries. However, it comes at the cost of having a +5.69% increase in mean X_0. It has a dead zone fraction (1.451%) which sits between Miura-ori's and Yoshimura's. Yoshimura's acceptance gain (+1.30 percentage points) sits between Miura-ori's and Kresling's on acceptance, but its radiation-length cost (+8.01%) and its dead-zone fraction (1.861%) are the highest of the three.

This makes Yoshimura hard to justify, especially when we compare it with Kresling. When compared directly to Kresling, it has a smaller acceptance rate and has the highest radiation-length cost and a higher dead zone fraction.

As pointed out in the fold angle sweep, Yoshimura's nominal fold angle (30°) operates in the lower-material regime of its parameter space; increasing θ yields higher acceptance but causes radiation length to escalate rapidly.

And so now, the most practical choice narrows down to Miura-ori or Kresling. And in this case, the winner depends on the requirement. A design that's acceptance-limited, where the dominant concern is closing dead zones and maximizing hit efficiency, favors Kresling, though this comes with more of the barrel's original dead-zone area relocated into concentrated overlap regions rather than removed outright. A design limited by radiation-length traversal, where multiple scattering from extra material is the dominant concern for momentum resolution, favors Miura-ori instead. This is because of its more uniform and barrel-like arrangement. Nothing in this dataset picks one of these over the other in general. It depends entirely on which constraint binds harder for a given application.

The thin-silicon evaluation at 50 μm (Section 4.8) reinforces these conclusions across sensor thicknesses. Reducing the silicon thickness from 300 μm to 50 μm decreases overall material traversal by $\sim 4.8\times$ while leaving geometric acceptance gains virtually unchanged (+2.37 pp for Kresling, +1.26 pp for Yoshimura, and +0.59 pp for Miura-ori). When compared against the bent, continuous sensor of ALICE ITS3 at matched physical scope, the flat-faceted tessellations retain a modest silicon-only material penalty (+0.010–0.016 pp in X/X_0) due to non-normal local incidence angles.

It is important to understand the impact of folding here, compared to the other techniques discussed. Folding doesn't eliminate the dead-zone/overlap trade-off. Here, we trade a real increase in acceptance for a real increase in mean radiation length and in the relocation of the dead-zones. The fold-derived tessellations sit somewhere between these — rigid flat facets like CMS's tilted modules, but with crease-line seams that neither of the other two approaches has to deal with. A fold pattern with a different unit cell than the three considered here could in principle land at a different point on this trade-off space; the three families tested are a sample of the design space, not an exhaustive one.

The quantitative findings reveal that origami folding does not magically circumvent the dead-zone/overlap dilemma; rather, it reshapes and redistributes the trade-off surface. For acceptance-critical applications where hermetic tracking coverage is paramount, **Kresling** is the preferred geometry (+2.36 pp acceptance gain), provided that tracking algorithms can handle localized material spikes at fold vertices. For precision momentum spectroscopy where Coulomb scattering dominates track errors, **Miura-ori** provides the optimal balance, delivering a reliable acceptance gain (+0.49 pp) with minimal material inflation (+1.88%) and uniform spatial response. **Yoshimura** is uncompetitive under standard cylindrical envelopes due to its excessive overlap seams.

6 Limitations and Future Work

No mechanical or structural analysis was done in this phase. All three fold geometries are evaluated purely on radiation-transport and dead-zone/hit-density grounds. Mechanical fabrication, structural support, and fold-vertex stress concentrations under load are outside the scope of this radiation-transport study.

The simulation used an idealized material assignment — silicon sensor layers and substrate only. Glue layers, wire bonds, mechanical supports, cooling infrastructure, and readout electronics weren't modeled. That was a deliberate scope decision and it was done because modeling supports, cooling, and electronics to a standard defensible enough to trust quantitatively would need its own set of references, its own material and geometry standardization, and its own independent validation for each subsystem — a scope of work established detector groups treat as a substantial undertaking in its own right, not a natural add-on to a geometry comparison like this one.

Based on the results above, the next steps are fairly direct. Mechanical and structural analysis of the fully deployed configurations, independent of the radiation-transport question addressed here, is a necessary next step to check whether the acceptance/radiation-length trade-off found here is compatible with a structure that can actually hold its shape under load. Physical prototyping of a Pareto-preferred geometry for a specific target application is the eventual goal, but remains outside the scope of this phase.

7 Conclusion

This project compared three cylindrically deployable origami fold patterns — Miura-ori [1], Yoshimura [2], and Kresling [3] — against a conventional flat-tile barrel detector of matching radius, using Geant4 [4] simulations of 5 GeV/c pions to evaluate hit acceptance, radiation-length traversal, and spatial dead-zone/overlap structure for each fully deployed geometry. A dedicated flat-plate model was used to validate the overall Geant4 setup.

Every folded geometry achieved a measurable acceptance gain over the flat barrel, but at the cost of increased radiation-length traversal. Kresling yielded the largest gain in particle acceptance (+2.36 pp), while Miura-ori produced the smallest acceptance gain but also the smallest radiation-length penalty (+1.88%). Yoshimura fell between the two in acceptance (+1.30 pp), yet exhibited the largest radiation-length increase (+8.01%) and dead-zone fraction of the three patterns. A fold-angle sweep further showed that the Yoshimura configuration studied here was already near the favorable region of its design space, indicating that its comparatively weaker performance was not simply a consequence of a suboptimal fold angle.

No single geometry is optimal across all metrics: Kresling is preferable when maximizing acceptance is the primary goal, whereas Miura-ori is preferable when minimizing additional radiation length is the higher priority. The appropriate choice therefore depends on which constraint dominates for a given detector application.

Furthermore, extending the simulation to a 50 μm thin-silicon regime (matching ALICE ITS3) confirms that the geometric acceptance advantages of folding are invariant under sensor thinning, while scaling total material traversal down by $\sim 4.8\times$. While folded planar facets cannot match the silicon-only material minimization of a continuous curved sensor, origami tessellations provide a viable and mathematically rigorous architectural framework for next-generation tracking detectors.

Overall, this study presents the first systematic Geant4 radiation-transport comparison of cylindrically deployable origami fold patterns against a planar-tiled detector baseline. Using a physically realistic, differentiated silicon/Kapton mesh, we show that folding consistently improves geometric acceptance at the expense of increased radiation-length traversal. Miura-ori and Kresling define distinct, highly attractive operating points on the Pareto frontier, establishing origami tessellation as a viable and compelling geometric design strategy for future deployable tracking detectors.

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A Full Multiple-Scattering Validation Dataset

Across all 20 points, $\sigma/\theta_{0,\text{Highland}} = 0.930 \pm 0.003$ (mean ± population standard deviation), ranging from 0.924 to 0.935; a weak downward trend against momentum does not reach conventional significance at this sample size (Section 2.6). The fitted tail index averages $\alpha = 2.999 \pm 0.046$ — indistinguishable from the Rutherford-predicted value of 3 to within fit uncertainty — while the tail's share of the total $\Sigma\theta^2$ averages 86.5% (range 81.9–89.7%) despite the tail itself comprising only 3.3–3.9% of events at any given point. Both properties are visualized directly in Figure 4.

Table 15: Full multiple-scattering dataset (all four geometries × five beam momenta, 20 points), Rayleigh-core-plus-Rutherford-tail mixture fit to the space-scattering angle (`scatterAngleDeg`), fit by unbinned maximum likelihood (Section 2.6). σ is the fitted Rayleigh core scale, directly comparable to Highland's projected θ_0 with no additional conversion factor; $\theta_{0,\text{Highland}}$ is the Highland-formula prediction (Eq. 2.9) at the same momentum and mean x/X_0; α is the fitted tail power index (Rutherford single-scattering prediction: $\alpha = 3$); tail fraction is the share of events beyond the fitted tail-onset boundary $\theta_{\min} = 3\sigma$. Binned χ^2/dof values are calculated over fine log-bins and reflect the high statistical sensitivity ($N \approx 4.2 \times 10^5$) of the full event sample.

Geometry	p (GeV/c)	N_{hit}	σ (deg)	$\theta_{0,\text{Highland}}$ (deg)	$\sigma/\theta_{0,\text{Highland}}$	Tail fraction	Tail index α	χ^2/dof (binned)
Barrel reference	0.5	415,207	0.07491	0.08008	0.935	3.29%	2.908	59.89
Barrel reference	1.0	415,763	0.03626	0.03894	0.931	3.44%	2.960	40.59
Barrel reference	2.0	415,408	0.01799	0.01933	0.931	3.50%	2.932	31.18
Barrel reference	5.0	415,794	0.00715	0.00772	0.927	3.57%	2.993	60.90
Barrel reference	10.0	416,043	0.00358	0.00386	0.928	3.65%	2.976	53.05
Kresling	0.5	426,932	0.07649	0.08211	0.932	3.68%	3.000	72.65
Kresling	1.0	427,408	0.03695	0.03995	0.925	3.89%	3.068	38.96
Kresling	2.0	427,092	0.01840	0.01983	0.928	3.81%	3.041	82.45
Kresling	5.0	427,547	0.00733	0.00792	0.925	3.93%	3.047	35.38
Kresling	10.0	427,690	0.00366	0.00395	0.927	3.88%	3.051	46.32
Miura-ori	0.5	418,117	0.07556	0.08081	0.935	3.33%	2.969	44.05
Miura-ori	1.0	418,508	0.03669	0.03930	0.934	3.45%	2.955	68.03
Miura-ori	2.0	418,312	0.01817	0.01951	0.931	3.51%	2.946	45.80
Miura-ori	5.0	418,451	0.00724	0.00779	0.930	3.50%	3.015	32.48
Miura-ori	10.0	418,057	0.00363	0.00389	0.932	3.50%	2.976	30.54
Yoshimura	0.5	421,642	0.07741	0.08315	0.931	3.63%	2.973	37.94
Yoshimura	1.0	422,099	0.03740	0.04046	0.924	3.76%	3.068	52.99
Yoshimura	2.0	422,310	0.01861	0.02008	0.927	3.81%	3.027	60.63
Yoshimura	5.0	422,587	0.00747	0.00802	0.932	3.71%	3.024	70.42
Yoshimura	10.0	422,058	0.00372	0.00401	0.927	3.74%	3.041	58.37

B Data and Code Availability

All baseline simulation datasets were generated under the differentiated-mesh Geant4 v11.x pipeline utilizing a nominal 1.0 mm 3D Gaussian vertex smear and 500,000 primary 500 MeV/c π^+ events per configuration at the fully deployed nominal state. Parameter sweep evaluations systematically varied facet count N, fold angle θ, and vertex smear σ around these baseline settings. Replication studies (Tables 5 and 9) utilize five independent simulation runs with distinct pseudo-random engine seeds.

The complete simulation source code, parametric geometry generators, automated pipelines, and figure reproduction scripts are openly available in the GitHub repository:

https://github.com/Alphacodde/Origami-Particle-Detectors-Geometries

The raw Geant4 ROOT ntuple datasets ($\approx$ 10.7 GB across all experiments and sweeps) are archived and accessible via Google Drive:

https://drive.google.com/drive/folders/1ZbLm0H2FCLPQepytwLGCHj7_VS4sdF78?usp=sharing

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